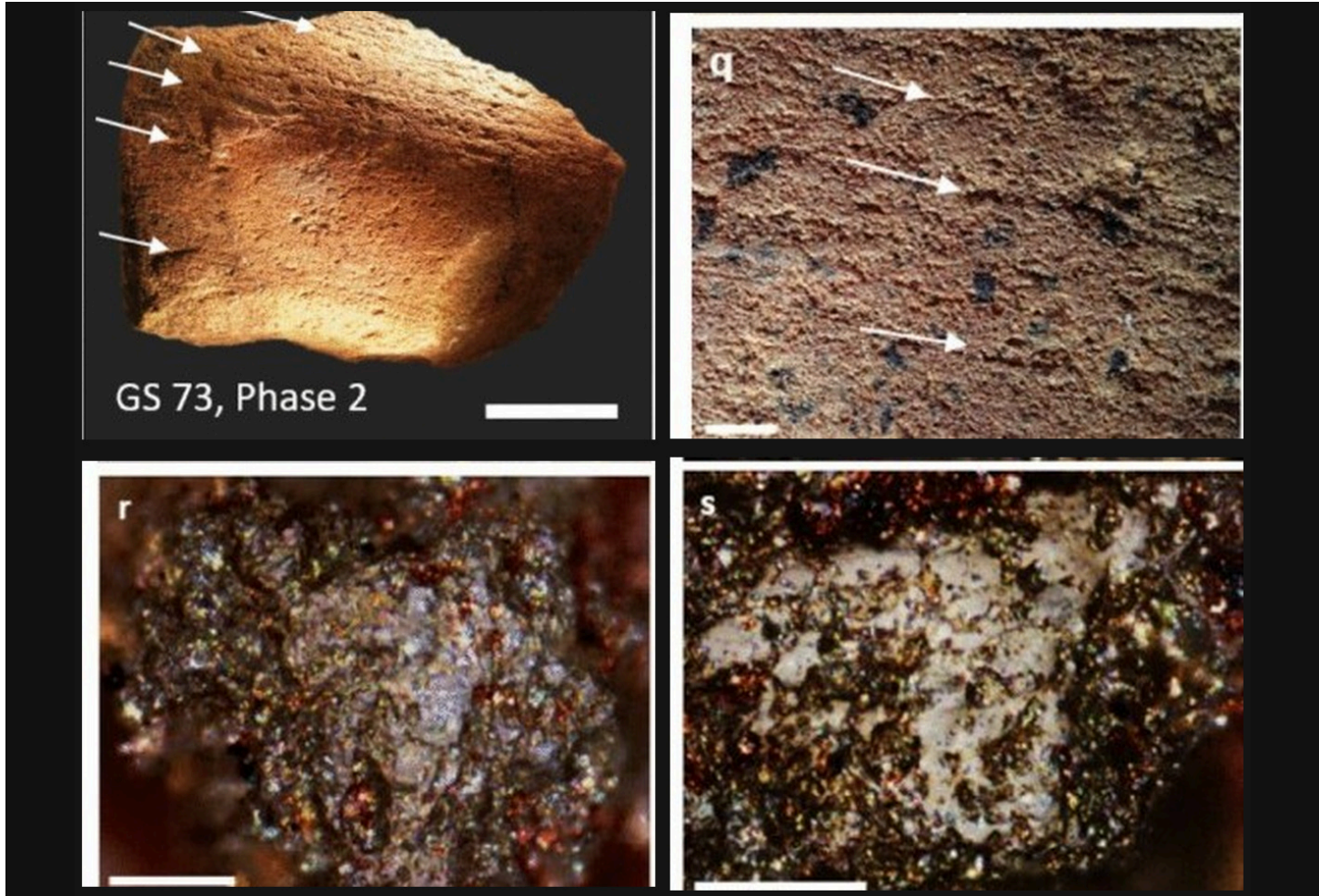


GS73

A broken piece of sandstone from a rock shelter in northern Australia, and the long journey by which it became evidence.



The four views above are all of the same object. The first is the artefact itself, photographed with a five-centimetre scale bar, with arrows marking the worn areas. The second is the same surface at low magnification, where striations and levelled quartz grains become visible. The last two are microscope views, taken at different points, showing bright polish on the raised parts of individual grains. Nothing has been added to the stone between one view and the next — only the magnification changes.

G S73 is a fragment of a millstone: the lower, fixed slab of a grinding pair, against which a smaller handheld stone is worked. It is sandstone, it is broken, and on its own it looks unremarkable. It was excavated from square B5/52 at the Madjedbebe rock shelter, on Mirarr Country in northern Australia.

Where a thing is found in the ground matters as much as the thing itself. GS73 came from Phase 2 of the excavation, a layer modelled to between 68,700 and 50,400 years ago. That range is not a measurement of the stone; it is a model of the deposit the stone was lying in, and the earliest end of it is disputed.

The stone reached a laboratory before it reached this page. Researchers photographed it, examined its surface under magnification, and washed and analysed what was still caught in it. They recovered 41 starch grains and detected plant-related compounds. They could not identify which plant. They also found two chemical peaks they identified as plastic contamination — a modern signal sitting inside an ancient sample, reported rather than quietly removed.

Hayes and colleagues read the combination — the shape, the deep partial grooves, the levelled grains and polish, the starch and the plant compounds — as evidence of plant processing, and published it in 2022 as part of an argument that grinding-stone use at Madjedbebe has a very long history. Several independent methods pointing the same way is what makes that reading strong.

The age of the layer is argued about separately. Clarkson and colleagues hold that the artefacts stayed where they were deposited, supporting occupation around 65,000 years ago. Williams and colleagues point to termite activity and sediment movement, and argue that the earliest material may not have stayed in place. Note what is and is not in dispute: the disagreement is about how old the layer is, not about whether GS73 is a genuine grinding stone.

None of this work was done on unclaimed ground. Madjedbebe is on Mirarr Country. The research was carried out with the permission of the custodians of Madjedbebe, May Nango and Djaykuk Djandjomerr, and of Mirarr Senior Traditional Owner Yvonne Margarula, with Gundjeihmi Aboriginal Corporation facilitating the work. The images and information were published for general educational purposes; no commercial use is authorised without prior Mirarr consent. Country and custodianship are part of this source’s context, not a footnote to it.

A single stone cannot tell you who used it, or what exactly they ground. What it can do — once it has been recorded, magnified, analysed, argued over and attributed — is carry evidence. That is the difference between an object and a source.

Source: panels adapted from Figure 3 of Hayes, E. H. et al. (2022), “65,000-years of continuous grinding stone use at Madjedbebe, Northern Australia,” *Scientific Reports* 12, 11747, doi:10.1038/s41598-022-15174-x (open access, CC BY 4.0). Figure prepared by Elspeth H. Hayes. The dating comparison uses Clarkson et al. (2017) and Williams et al. (2021). **The research was conducted with permission from the custodians of Madjedbebe, May Nango and Djaykuk Djandjomerr, and Mirarr Senior Traditional Owner Yvonne Margarula, with Gundjeihmi Aboriginal Corporation facilitating the work.** Mirarr-related images and information were published for general educational purposes; no commercial use is authorised without prior Mirarr consent. See SOURCE-NOTICE.md.